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From Equity Pairs to Options Relative Value

Can learning how much to trade improve a volatility strategy?

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Abstract

Pairs trading starts with a simple idea: two related securities can become unusually expensive or cheap relative to each other. We apply that idea to options, where the price of uncertainty can differ across related stocks. We first search for a trading signal and then ask whether a learning algorithm can improve the amount of risk taken on each trade. Our strongest result uses roughly one-month options on the downside of the price distribution. In a simulated $10 million account from January 2018 through August 2025, learned sizing produces an annualized Sharpe ratio of 1.003, compared with 0.769 for constant full exposure. Both figures measure returns above a cash benchmark. This is a promising historical comparison, although the strategy was selected after many experiments on these dates.

1. Why trade volatility rather than stock prices?

Suppose AMD and NVDA are exposed to similar news, but AMD options suddenly become much more expensive relative to NVDA options than they usually are. An investor could sell AMD options and buy NVDA options, hoping that the difference is excessive. The trade concerns the prices of the options and the subsequent stock movements. It does not require a forecast that AMD shares will fall or NVDA shares will rise.

This distinction matters because an option is affected by both the direction and the size of stock-price moves. We use stock positions to offset the options' immediate sensitivity to direction. We also match their initial sensitivity to changes in implied volatility. The resulting position is designed to compare volatility across two names, although it still carries risk when their prices, volatility or hedge requirements evolve differently.

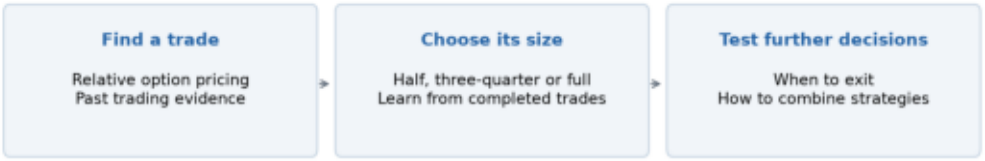


Figure 1. The two main decisions: identify a relative-price opportunity, then choose exposure. Management and portfolio allocation are tested separately.

Our question has two parts. First, can observable differences in option pricing identify useful trades after execution and hedging costs? Second, given those trades, can an algorithm decide when to take more or less exposure? Keeping these questions separate helps us understand where an improvement comes from. A stronger signal and a better sizing policy are different contributions.

We build on Ning and Lee's study of reinforcement learning for equity pairs [1]. Their paper supplies the starting idea of learning how to trade a relative-price signal. Our extension changes the instrument, the accounting and the decision being learned. The following pages explain the option trade before describing that connection, our experiments and the evidence for adaptive sizing.

2. What the option trade actually does

A put gives its buyer the right to sell stock at a specified price, called the strike, until the option expires. A put with a strike below the current stock price is out of the money. Our retained strategy selects these puts near 25 delta and approximately 30 days to expiry. “25 delta” means that a small $1 rise in the stock would initially lower the put’s value by about $0.25 per share. It is a measure of price sensitivity, rather than an exact probability of exercise [2].

Implied volatility (IV) is the volatility input that makes an option-pricing model reproduce a quoted option price. A higher IV generally means a more expensive option, holding other inputs fixed. Realized volatility (RV) describes how much the stock actually moved over a period. Selling an option can be attractive when its price more than compensates for those future movements; buying can be attractive in the opposite situation. Future RV is unknown at entry, so any estimate must use information already available.

A worked example: matching vega and hedging delta

Vega measures how much an option’s value changes when IV rises by one percentage point—for example, from 30% to 31% [3]. Consider the illustrative positions below. Vega is stated per standard 100-share contract; these are explanatory numbers, not an observed trade.

Position	Option contracts	Vega per vol point	Initial stock hedge
Buy A puts	+100	+$450	Buy 2,500 shares
Sell B puts	−150	−$450	Short 3,750 shares

Each A contract has $4.50 of vega and each B contract has $3.00. Buying 100 A contracts and selling 150 B contracts therefore balances +$450 against −$450. A common one-point IV increase would approximately cancel at entry. Both puts initially have delta −0.25 per share: the long puts require buying stock, while the short puts require selling it. We adjust these stock hedges daily as delta changes. Equal contract counts would generally fail to balance vega.

If A’s IV instead rises by two points while B’s rises by one, the initial vega approximation gives a $900 gain on A and a $450 loss on B: a $450 net gain before other effects. The hedge removes an equal common IV move, while preserving exposure to the difference. This approximation describes a small change in option marks. It is not a formula for the profit earned by holding the pair to expiry.

What earns or loses money?

After delta hedging, option returns still depend on the stock’s path. Gamma describes how delta changes as the stock moves. Hedging a long put buys more stock after declines and sells after rises; the short holder does the opposite. The long option’s gains from movement must compensate for time decay and costs. Changes in IV also affect the interim value [4]. If held to expiry, each leg’s result comes from its purchase or sale price, terminal payoff, stock hedges and costs.

Our actual-contract backtest records these cash flows. It does not replace them with “realized volatility minus implied volatility.” Initial vega neutrality also does not make the position riskless: the two legs can have different gamma, react differently to market stress, and drift apart as time passes. We match vega when entering; we do not continuously rebalance it.

3. What we take from the motivating paper

Ning and Lee [1] study stock-price pairs. They form a spread by subtracting a weighted price of one stock from the other, then trade deviations from its usual level. If a spread that has risen unusually far tends to fall back, selling it can profit from mean reversion. Their empirical mean reversion time measures how long historical departures from the mean took to reverse. They choose spread weights partly to shorten that time.

The trading decision is handled with Q-learning. The state summarizes recent spread movements; the action changes the trading position. A Q-value estimates the reward from an action together with the value of subsequent decisions. That continuation value is what makes the problem sequential: holding a position today changes what can be earned or done tomorrow. The learner trains on simulated mean-reverting spreads before being applied to observed stock prices.

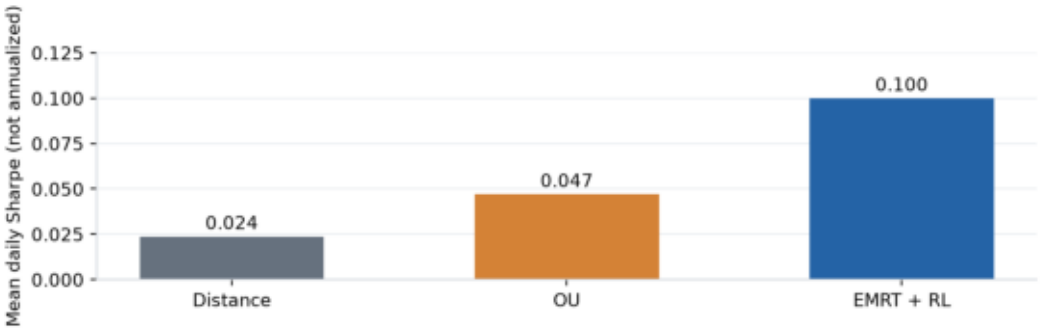


Figure 2. Mean daily Sharpe across the motivating paper's ten equity pairs. Their daily metric and equity setup differ from our annualized options results; the values should not be compared directly.

What the evidence establishes—and leaves open

The paper forms ten equity pairs using 2022 data and evaluates trading in 2023. Its combined method performs better on its reported daily Sharpe measure than the two comparison methods. However, it changes both the spread construction and the trading policy. The result supports that combination, but cannot tell us how much improvement came from Q-learning alone. A comparison using identical spreads and changing only the trading policy would answer that narrower question.

The reward used in training also depends on a known long-run mean. That mean is available in the synthetic model, whereas a real spread's equilibrium can shift after changes in business conditions. Rapid historical reversion need not imply profitable future reversion. One test year leaves this transfer especially difficult to judge, and correlated pairs should not be treated as independent evidence.

We carry forward the separation between finding a spread and learning how to trade it. We change the target from stock-price convergence to the recorded profit of a delta-hedged option position. Our main learner chooses entry size; it does not use a Q-table or learn a sequence of exits. We later test a closer Q-learning adaptation for position management. These are distinct experiments, and a successful sizing result does not establish that the paper's original algorithm transfers to options.

4. Data and a day in the backtest

We obtain historical option quotes and underlying-stock data through WRDS, together with rates and dividends needed for accounting. Early research used interpolated at-the-money IV: a convenient summary of option prices near the stock price. It does not necessarily identify a contract we could buy. The reported strategy instead selects actual puts with their own strikes, expiry dates, quotes and hedge sensitivities. We fit IV and hedge sensitivities using Black–Scholes, a transparent pricing approximation whose European exercise assumption is a limitation for American equity options.

The research roster contains 345 candidate pairs. Available contracts narrow the prepared 25-delta panel to 125 pairs, 161 assets and 36,327 dated candidate observations. Further signal screens produce 608 proposals during evaluation; 42 pairs ultimately trade under learned sizing. These numbers describe successive stages, rather than 345 continuously tradable pairs. Missing or unusable quotes matter, especially when comparing different parts of the option curve.



Figure 3. A proposal uses the signal-date information and contract identities. Subsequent fills, stock hedges and settlement determine the actual profit.

From information to inventory

At a session’s close, we select options near the target delta and expiry and calculate the signal using information through that date. A qualified proposal fixes which contract to buy and which to sell. Execution takes place in the next session using those contract identities. The chosen vega budget determines integer contract counts, with no more than 10% initial vega mismatch. Quote checks and account limits can reject an order.

Once entered, the stock hedge is adjusted daily. The retained strategy holds each option to its own expiry, except for audited corporate-event exits. The simulator tracks option and stock cash flows, dividends, cash interest and borrowing in one account. Stock hedges are netted when multiple trades use the same name. This shared ledger matters: overlapping trades consume capital and can affect whether a later proposal is admitted.

Execution and capital assumptions

Option fills use the quoted midpoint with an adverse penalty equivalent to 0.10 IV percentage point on each leg. This is a volatility unit, not ten cents or 10% of the option price. Evaluation also charges $0.03 per contract and one basis point of stock-hedge turnover. Training outcomes retain a higher $0.65 commission. The assumed fills are not verified live executions.

Cash earns the benchmark rate; negative cash incurs that rate plus 1%, and short stock has a 0.3% annual borrow charge. Entry checks impose a 5% stress limit, an 80% reserve limit and a gross stock-hedge limit of twice account equity. Stress estimates potential losses under large moves; reserves approximate capital tied up by open positions. These are inherited research assumptions, not broker margin requirements. Early assignment, market impact and forced liquidation are not fully modeled. We use the same $10 million account and limits for the main sizing comparisons.

5. Finding a signal before learning its size

The early ATM proxy results gave us reason to investigate further, but they weakened when we simulated actual contracts. Strike-specific payoffs, daily hedge errors and costs were absent from the simple IV-versus-future-RV proxy. We then investigated maturity, location on the option curve and additional predictors. Roughly 30-day, 25-delta puts were the most useful retained configuration. We found this configuration through research; it was not a choice made before seeing the results.

The direction rule

For each pair, we compare the IV difference with its preceding 252 trading sessions. A current difference in the lowest or highest fifth of that history can propose a trade. At the high extreme, the rule buys the relatively cheaper name's put and sells the richer name's put. Historical ranking matters because related companies can have persistently different volatility. An IV difference of ten points means something different for a pair usually five points apart than for a pair usually fifteen points apart.

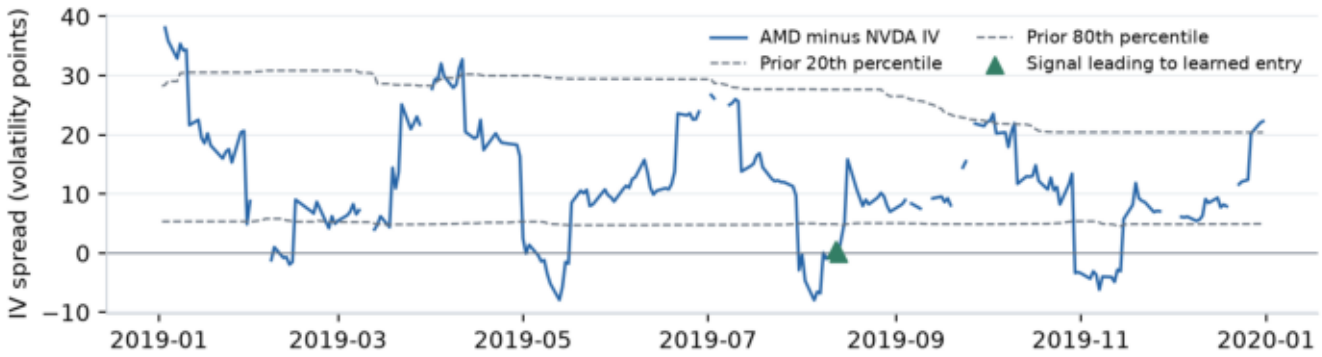


Figure 4. AMD minus NVDA put IV, 2019, with boundaries calculated from prior observations. Green markers show signals that lead to learned-policy entries in the following session. Gaps indicate unavailable quotes.

Figure 4 illustrates the observable spread and its changing historical boundaries. Each day's series uses the contract selected for that day. It is therefore a signal series, not the mark of one option held throughout the year. After an entry, the backtest follows the purchased and sold contracts without replacing them as the daily selection changes.

Why an extreme is not enough

We require a return beta between 0.85 and 1.15 and return correlation of at least 0.5. Beta measures the scale of one name's return response to the other; correlation measures how consistently they move together. Neither guarantees that their option prices will behave alike. These screens keep the comparison among reasonably related names without treating relatedness as the trading signal itself.

The retained strategy also requires positive directional profitability over the preceding 91 and 365 days, with at least 21 and 63 completed observations respectively. This memory uses simulated prior candidate trades whose outcomes have already finished. An option entered yesterday with a month left cannot contribute its eventual profit today. The requirement seeks directions with recent evidence of profitability, but it is itself a selected rule. Page 9 tests removing both requirements while keeping memory as an input to learned sizing.

6. What information should determine exposure?

A wide IV spread can reflect genuinely different future uncertainty. We therefore give the learner information about the spread, recent stock movements and the trading environment. Baseline inputs include historical spread rank, IV and RV levels, beta and correlation, days to expiry, VIX as a measure of broad market volatility, expected costs and completed profitability. They describe a trade at decision time; they do not include its future payoff.

Estimating relative richness

The most useful added block compares implied variance with a simple estimate of future variance. We estimate the latter from exponentially weighted squared daily returns, with a 20-session half-life: an observation's weight halves over twenty sessions. At least 63 observations are required. Multiplying by 252 puts the estimate on an annual basis. The difference between squared annual IV and forecast annual variance is our estimated volatility risk premium.

Estimated variance premium = IV^2 − forecast realized variance

For example, IV of 40% and forecast RV of 30% imply a premium of $0.40^2 - 0.30^2 = 0.07$. This is variance, not seven volatility points. We compare the premium across the proposed long and short. A historical z-score subtracts each name's prior average and divides by its prior variability, distinguishing unusual richness from a normal difference. We also include a forecast-to-implied variance ratio. The retained policy has 33 inputs, including fields that indicate when a value is unavailable.

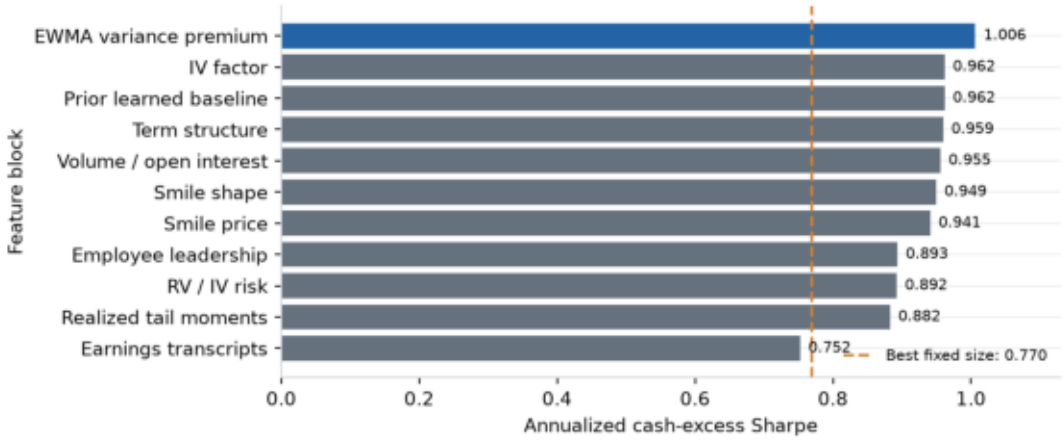


Figure 5. Development comparisons: one feature block added at a time. These use the original $345 million research account; the controlled $10 million comparison follows on page 8.

Figure 5 compares adding each feature block to the same development setup. Estimated variance premium raises Sharpe from about 0.962 to 1.006; several more elaborate inputs reduce it. These scores use the original $345 million development account, so the 1.006 is not the $10 million headline result. We have evidence that this block helps allocation, rather than evidence that our simple RV forecast is optimal.

Training history also matters. Refitting monthly on 18 months of completed outcomes produced a development Sharpe of 0.962, compared with 0.826 for five years and 0.786 for six months, before adding the premium block. Short history adapts quickly but supplies few examples; long history mixes regimes. We retained 18 months after these comparisons. Both window length and feature selection were informed by the evaluation history.

7. How the learner chooses a trade size

The entry rule supplies the direction and the option contracts. The learner's action is 50%, 75% or 100% of a full vega budget of $450 per IV point on each leg. Half size targets $225 per leg. These are exposure targets; actual contract counts depend on the selected options. Every action is positive, so this retained policy cannot choose to skip a qualifying trade. An account or quote check can still prevent execution.

States, actions and the objective

A state is the information available when sizing a proposal: the inputs described on the previous page. A regression tree groups similar states into at most eight leaves. Each leaf receives one of the three sizing actions, so a policy is a mapping from observed conditions to exposure. The grouping makes a manageable decision problem from a limited history of completed trades.

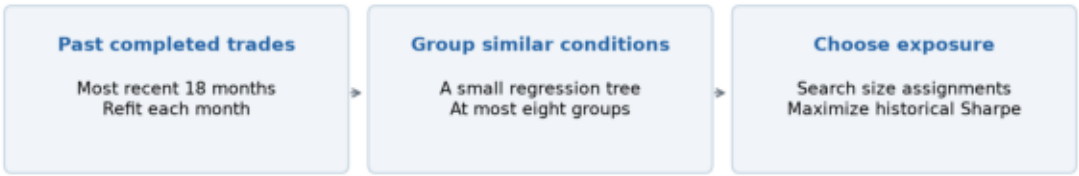


Figure 6. The monthly fit groups current conditions and searches for a size assignment with a better combined historical Sharpe. Only completed outcomes enter training.

We search over these mappings using the cross-entropy method, a direct policy-search algorithm. It generates candidate size assignments, scores them on the combined daily profit paths of completed historical trades, keeps the better assignments and uses them to guide the next round. Each fit uses 50 rounds with 192 candidates per round. The objective is portfolio Sharpe, which rewards profit relative to variability; it is not classification accuracy or the number of winning trades.

This approach learns from the economic reward of taking an action, which is why we discuss it as contextual reinforcement learning. It does not estimate a Q-value or a discounted continuation value. It also does not train a complete inventory-management agent inside the funded simulator. Historical size assignments are optimized first; the resulting policy is then replayed with shared cash, inventory and entry limits.

What is known at each refit?

For a January 2018 decision, the fit uses completed outcomes from the preceding 18 months, rather than the profit that January's options will later generate. The policy is refitted monthly. As evaluation advances, completed earlier evaluation trades become eligible for later training. This is a walk-forward procedure: each fit precedes the decisions it controls.

The first fit contains 74 completed examples. Its optimized historical score is 2.015 versus 1.573 at constant full size. That is an in-sample score on aggregated training paths, not the Sharpe of a funded account before 2018. It shows that the optimizer improves its training objective. Whether that improvement carries forward must be judged from subsequent trading, while remembering that the research team has inspected those subsequent dates repeatedly.

Walk-forward fitting prevents a monthly model from using unfinished future outcomes. It does not remove research selection: we chose signals, windows, strikes and policies after inspecting these dates, and pair formation is retrospective. A prospective universe and an untouched later period remain necessary to test whether the measured improvement survives.

8. The controlled sizing comparison

We replay three policies on the same 608 proposed trades from January 2018 through August 2025: constant half size, constant full size and learned sizing. Signal rules, proposed directions, contract identities, costs and account limits are shared. The question is whether conditioning size on the observed state improves the funded strategy relative to these constant-exposure controls.

Policy	Excess Sharpe	Trades entered	Profit above cash
Always half size	0.761	263	$248,315
Always full size	0.769	133	$272,338
Learned sizing	1.003	212	$322,637



Figure 7. Cumulative growth relative to the cash benchmark in the $10 million account. Proposals are shared; position sizes and executed portfolios differ.

What the comparison says

Learned sizing increases annualized excess Sharpe from 0.769 at full size to 1.003, a gain of 0.234. Profit above the cash benchmark rises from $272,338 to $322,637. Learned sizing enters 212 trades, compared with 133 at full size and 263 at half size. Smaller positions sometimes leave room for proposals that full-size inventory would block. The result therefore includes the effects of size on admission; it is not a comparison of identical executed trades multiplied by different constants.

Why cash must be separated from trading profit

Most of the $10 million account remains in cash. The learned policy’s funded annualized return is 2.963%, while its annualized growth relative to the cash benchmark is 0.342%. Reporting the funded return alone would give the options strategy credit for interest it could have earned without trading. We calculate Sharpe from daily portfolio returns minus daily cash-benchmark returns:

Annualized excess Sharpe = √252 × mean(daily excess return) / standard deviation(daily excess return)

These figures support adaptive sizing within the retained strategy. They do not prove a general advantage over supervised learning, which is not the control here. A block bootstrap gives a 95% interval of roughly −0.031 to +0.416 for the Sharpe improvement over full size. The interval includes zero, and does not account for selecting this strategy after many research trials. The point estimate is encouraging; its uncertainty remains material.

9. Participation, capital use and uneven returns

Figure 8 separates total returns from returns above cash. The retained strategy's strongest year is 2022: 1.676 percentage points above cash and Sharpe 2.462. In 2024, total return is 5.149%, but the account trails cash by 0.238 percentage point: Sharpe −0.785. Trading also trails cash in 2019 and 2021.



Figure 8. Learned sizing in the $10 million account: annual returns and annualized excess Sharpe. Return above cash is measured in percentage points. 2025 is partial through 29 August.

What happens if the profitability screens are removed?

Removing both memory requirements from page 5 expands proposals from 608 to 4,744. Memory remains a sizing input; spread, beta and correlation screens, contract selection, hedges, costs and account limits remain unchanged. Monthly fits use the broader set of completed outcomes. This expands eligible trades rather than simply enlarging existing positions.

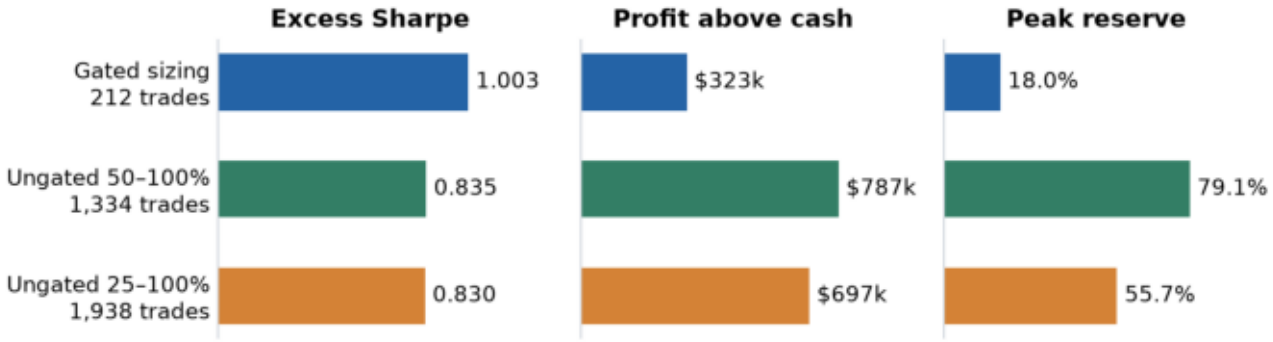


Figure 9. Same $10 million account and dates; memory screens change eligibility. Reserve is gross option premium plus half of gross netted stock-hedge notional. Figures show peak share of equity, not average deployment or broker margin.

Ungated sizing enters 1,334 trades across 77 pairs and earns $786,690 above cash, versus 212 trades across 42 pairs and $322,637 retained. Sharpe falls from 1.003 to 0.835; peak reserve rises from 18.0% to 79.1%, and maximum drawdown relative to cash rises from 0.49% to 1.03%. Broader participation increases profit and capital use at lower risk-adjusted performance.

A 25% sizing action gives 0.830 Sharpe and 1,938 trades, with lower profit and reserve. Ungated constant full sizing scores 0.713. Alternative action menus, state counts and training windows do not improve 0.835. Ungated Sharpe is roughly zero in 2024–2025: more trades do not establish steadier performance or independent evidence.

10. Extensions and what we learned

Longer-dated experiments also test when to reduce or close positions. Exiting can avoid losses, surrender future hedging gains and incur another spread. Good entry selection alone does not establish good management.

Experiment	Control Sharpe	Learned alternative
30-day puts: optional management	1.003 hold	1.029, 36-month fit
30-day puts: fitted-Q exits	1.003 hold	0.598, same 212 entries
180-day puts: conservative Q	0.592 hold	0.610 managed
365-day puts: new entry signal	0.254 fixed half	0.313 learned size

Compare policies within each row: maturities, signals and training procedures differ across rows. The 30-day fitted-Q experiment learns continuation values from actual profit rather than the motivating paper's known-mean reward. It weakens performance despite preserving the original entries. Longer options benefit modestly from selected experiments, but entry sizing and management do not consistently reinforce each other.

Combining saved 30-, 180- and 365-day returns gives Sharpe 1.006 at equal weights, and 0.865 with weights based on the preceding two years. This combines saved returns without a new shared-account replay. Losses occurring together can erase individual policy benefits. Buying cheap one-year vega without an offsetting short remains unprofitable after costs; management reduces losses without making it profitable.

The strongest finding is specific: conditional sizing improves this researched downside-option strategy relative to constant exposure. Removing profitability screens increases participation and total profit at a lower Sharpe. Neither variant offers steady returns in every year, and overlapping trades share names and market shocks. The broader sizing gain over full size has an unadjusted bootstrap interval crossing zero. Prospective proposals and an untouched evaluation period would provide stronger evidence; learning exposure across separately validated signals is a further portfolio question.

A 25-delta put trades downside volatility, like a 75-delta call at the same strike and expiry. Matched-contract checks found similar pre-cost profits; quotes, execution and hedge financing affect net differences. These effects should be checked before claiming a separate signal.

References and supporting material

[1] Ning, B. and Lee, K. (2024). Advanced Statistical Arbitrage with Reinforcement Learning. arXiv:2403.12180. https://arxiv.org/abs/2403.12180

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Data, settings, full experiment tables and figure sources accompany the repository: github.com/kreitzz/rl-atm-vega-pairs. The retained specification is config/p25_submission.json. paper/source_manifest.json records the source hashes. This report uses saved results; no models were refitted for its preparation. Academic use only; vendor data retain their institutional license restrictions.